AIM:- RNN Forward pass with Fixed Weights

```
import numpy as np
def softmax(x):
    exp_x = np.exp(x - np.max(x))
    return exp_x / np.sum(exp_x)
def cross_entropy(y_true, y_pred):
    return -np.sum(y_true * np.log(y_pred + 1e-12))
Wxh = np.array([
        [-1.753416, 0.9836007, 1.2459822],
        [-1.117427, 2.0291977, 0.5590203]])
Whh = np.array([[0.3323058, 1.704173],
                [0.3323068, 1.704173],
                [-0.4059835, 1.069699]])
Why = np.array([[5.0668307, 1.239699],
                [-3.2882083, -2.307849],
                [-0.8786225, 2.268150]])
bh = np.array([[-0.0238333], [0.1707909]])
by = np.array([[0.03411441], [-0.21828305], [0.78416864]])
x1 = np.array([[1], [0], [0]])
x2 = np.array([[0], [1], [0]])
x3 = np.array([[0], [0], [1]])
inputs = [x1, x2, x3]
t1 = np.array([[0], [1], [0]])
t2 = np.array([[0], [0], [1]])
t3 = np.array([[1], [0], [0]])
targets = [t1, t2, t3]
h0 = np.array([[0], [0]])
hs = [h0]
ys = []
ps = []
loss = 0
for t in range(3):
    xt = inputs[t]
    Wxhxt = Wxh @ xt
    print("Wxhxt")
    print(Wxhxt)
    Whhhs = Whh @ hs[t]
    print("Whhhs")
    print(Whhhs)
```

```
combind = Wxhxt + Whhs + bh
print ("combined")
print (combind)
ht = np.tanh(combind)
print ("ht")
print (ht)
y = Why@ht + by
print ("y")
print ( y)
p = softmax(y)
print ("p")
print(p)
hs.append(ht)
ys.append (y)
ps.append (p)
print ("current prediction (input):")
print (xt)
print ("next Prediction (probabilities):")
print ( np.round(p, 1))
loss_entropy = cross_entropy (target [t], p)
print("loss_entropy")
print ( loss_entropy)
loss += loss_entropy
print ("Total Loss :", Loss)
print ("Hidden states:")
for i, h in enumerate(hs):
        print (f"h{i}:")
        print(h)
```

output

```
Wxhxt
[[-1.753416]
 [-1. 1174217]]
Whhs
 [[0.] [0.])
combind
 [[-1.7772493]
  [-0.9466361]]
ht
 [[-0.94439854]
  [-0. 73825635]]
```

```
y
[[-5.66620879]
 [4.59088026]
 [-0.06053769]]
p
[[3.47744867e-05]
 [9.90507804e-01]
 [9.45742167e-03]]
current prediction (input):
 [[1]
  [0]
  [0]]
next prediction (probabilities):
 [[0.] [1.] [0.]]
loss_entropy
0.009537354183677206
Wxhxt
[[0.9836007] [2.0291977]]
Whhhs
 [[-1.57194565] [-0.40630185]]
combind
 [[-0.61217825] [1.79368675]]
ht
 [[-0.54565861] [0.9461483]]
y
[[-1.55770628] [-0.60761129] [3.40960284]]
p
[[0.00679229] [0.01756455]] [0.97564316]
current prediction (input):
 [[0] [1] [0]]
next prediction (probabilities):
 [[0.] [0.] [1.]]
loss_entropy
0.024658373663724358
Wxhxt
 [[1.2459822] [0.5590203]]
Whhhs
 [[1.43107487] [1.2336 2228]]
combind
 [[2.65322377]
  [1.96343348]]
```

ht
[[0.99012992] [0.96135096]]

y
[[6.24272093] [-5.69268934] [2.09470644]]

p
[[9.84443527e-01] [6.45220247e-06] [1.55500213e-02]]

current prediction (input):

[[0] [0] [1]]

next prediction (probabilities):

[[1.] [0.] [0.]]

loss entropy
0.015678745140 60932

Total Loss: 0.04987465298 7910884.

Hidden states:

h0:
[[0][0]]

h1:
[[-0.94439854] [-0.73825635]]

h2:
[[-0.54565861] [0.94461483]]

h3:
[[0.99012992] [0.96135096]]